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Assignment No.05

1. Introduction –

The application allows users to input a starting location and a destination. Upon submitting the information, it opens Google Maps to display the directions between the two locations. The user initiates the process by clicking a button to retrieve the directions.

2. Tools and Technologies used -

Tools and Technologies used in this Project are Android Studio, SDK compiler.

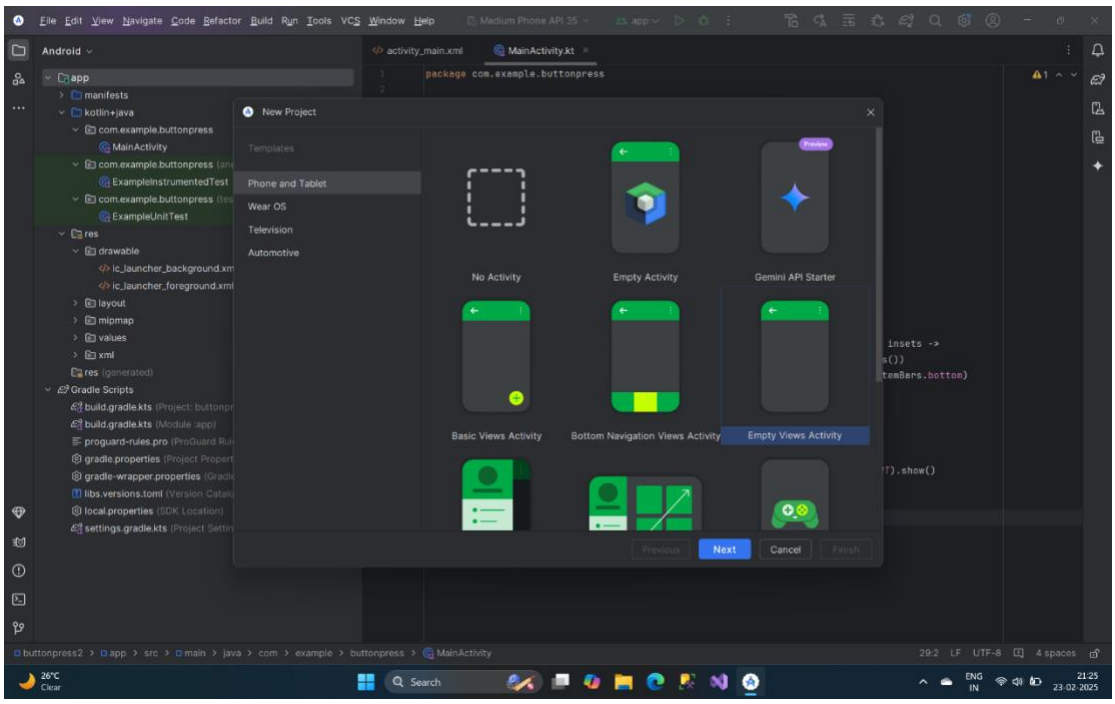
Mobile Emulator used - Medium phone API 35.

Language used – Java

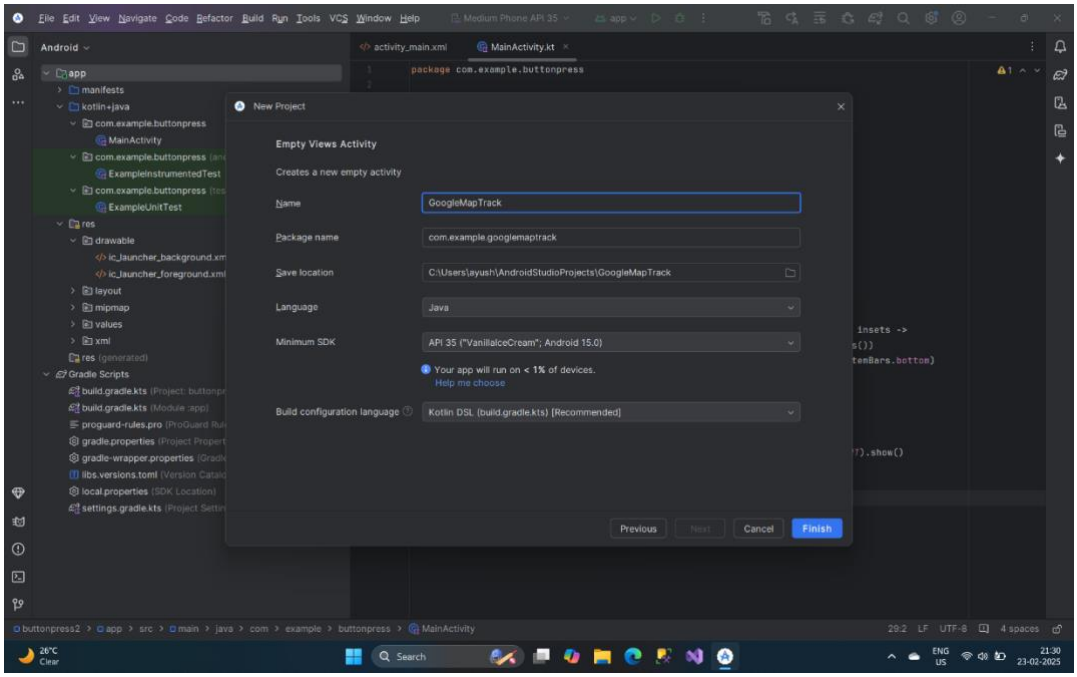
3. Procedure and steps –

The following steps were followed to complete the given task.

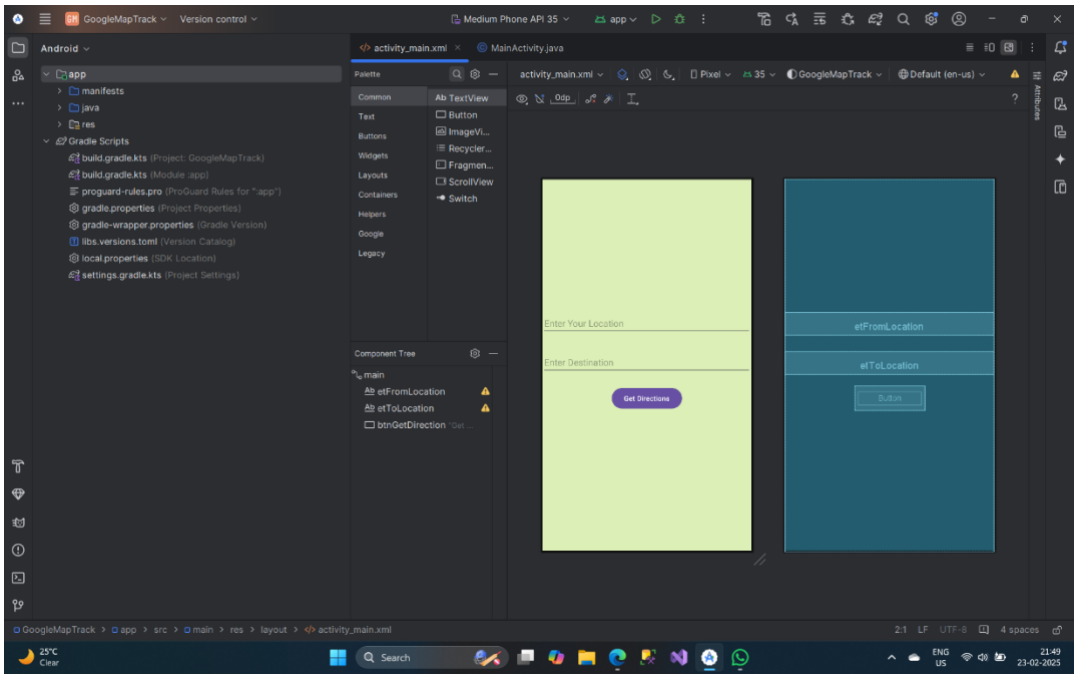
Step 1: Create a New Project:



Step 2: Naming and selecting language:



Step 3: Designing the UI :



Step 4: Writing the Code of xml:

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
```

```
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#DBEFB6"
    tools:context=".MainActivity">

    <EditText
        android:id="@+id/etFromLocation"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:ems="10"
        android:hint="Enter Your Location"
        android:inputType="textPersonName"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.38"
        tools:ignore="Autofill,HardcodedText,TouchTargetSizeCheck" />

    <EditText
        android:id="@+id/etToLocation"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_marginTop="32dp"
        android:ems="10"
        android:hint="Enter Destination"
        android:inputType="textPersonName"
        app:layout_constraintEnd_toEndOf="@+id/etFromLocation"
        app:layout_constraintStart_toStartOf="@+id/etFromLocation"
        app:layout_constraintTop_toBottomOf="@id/etFromLocation"
        tools:ignore="Autofill,HardcodedText,TouchTargetSizeCheck" />

    <Button
        android:id="@+id/btnGetDirection"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="24dp"
        android:text="Get Directions"
        app:layout_constraintEnd_toEndOf="@id/etToLocation"
        app:layout_constraintStart_toStartOf="@id/etToLocation"
        app:layout_constraintTop_toBottomOf="@id/etToLocation"
        tools:ignore="HardcodedText" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

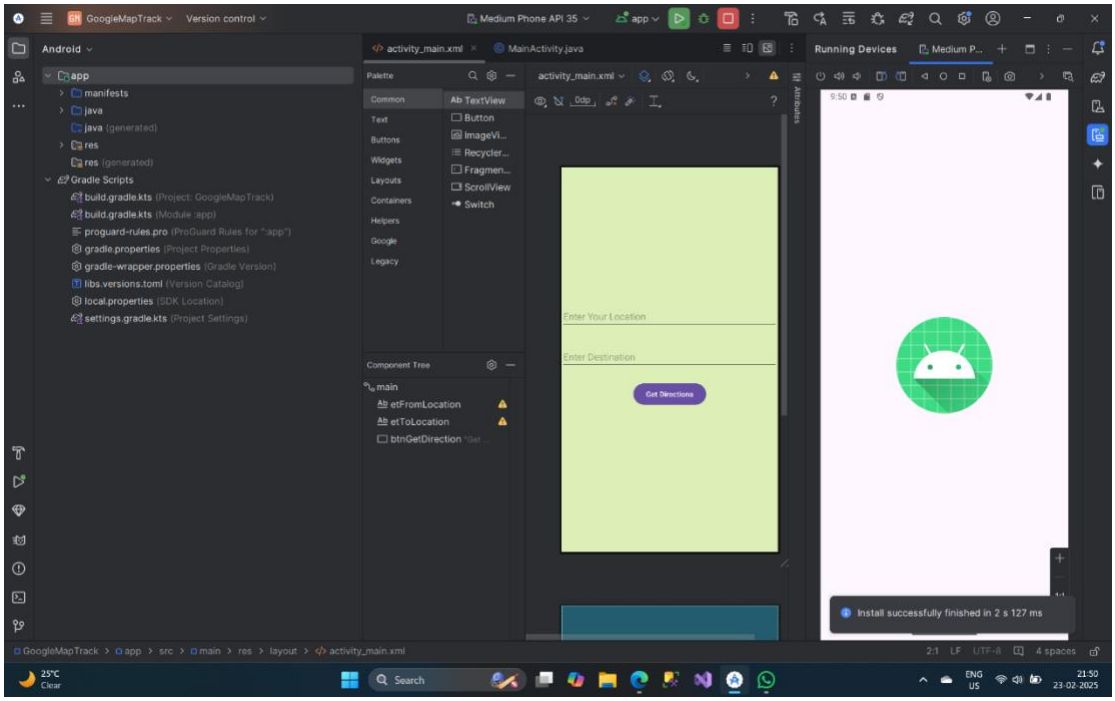
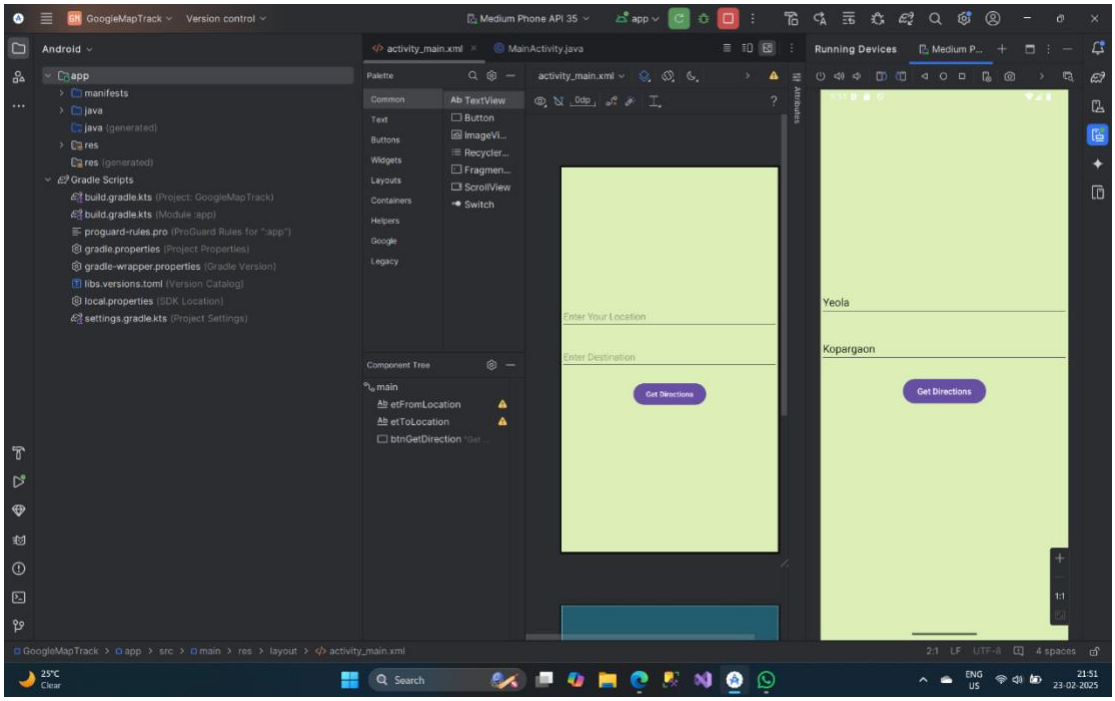
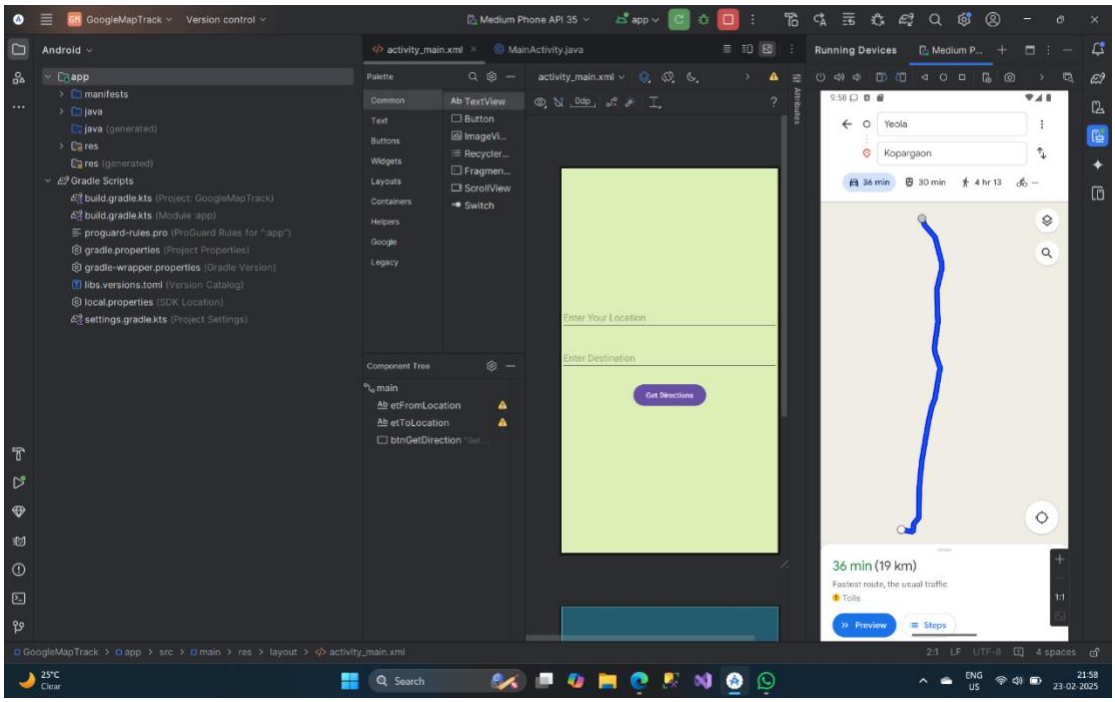
Step 5: Writing code of mainactivity.kt:

```
package com.example.googlemaptrack;

import android.content.ActivityNotFoundException;
import android.content.Intent;
import android.net.Uri;
import android.os.Build;
```

```
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;

import androidx.annotation.RequiresApi;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private EditText etFromlocation;
    private EditText etTolocation;
    private Button btnGetDirection;

    @RequiresApi(api = Build.VERSION_CODES.S)
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etFromlocation = findViewById(R.id.etFromLocation);
        etTolocation = findViewById(R.id.etToLocation);
        btnGetDirection = findViewById(R.id.btnGetDirection);

        btnGetDirection.setOnClickListener(v -> {
            String userLocation = etFromlocation.getText().toString();
            String userDestination = etTolocation.getText().toString();

            if (userLocation.isEmpty() || userDestination.isEmpty()) {
                Toast.makeText(this, "Please enter both locations",
Toast.LENGTH_SHORT).show();
            } else {
                getDirections(userLocation, userDestination);
            }
        });
    }

    private void getDirections(String from, String to) {
        try {
            Uri uri = Uri.parse("https://www.google.com/maps/dir/" + Uri.encode(from) + "/" +
Uri.encode(to));
            Intent intent = new Intent(Intent.ACTION_VIEW, uri);
            intent.setPackage("com.google.android.apps.maps");
            intent.setFlags(Intent.FLAG_ACTIVITY_NEW_TASK);
            startActivity(intent);
        } catch (ActivityNotFoundException exception) {
            Uri uri =
Uri.parse("https://play.google.com/store/apps/details?id=com.google.android.apps.maps");
```

```
                Intent intent = new Intent(Intent.ACTION_VIEW, uri);
                intent.setFlags(Intent.FLAG_ACTIVITY_NEW_TASK);
                startActivity(intent);
            }
        }
}
```

Step 6: Running the application on Emulator



Step 7: Output: